

LESSON 3-5 • SESSION 1 • PRACTICE • FORM B

Comparing Ratios with Common Terms

Learning target – I can compare two ratios by scaling them until they share a common term. • 6.RP.A.3b

REMEMBER

- To compare two ratios, **scale** them until one term is the **same** in both. Use a common multiple.
- Then compare the **other** term. Less water for the same soap means a **stronger** mix.
- Never compare the raw numbers: $3 : 7$ vs $2 : 5$ is not settled by $7 > 5$.

WORKED EXAMPLE

Mix A has 2 soap to 5 water. Mix B has 3 soap to 7 water. Which is stronger?

Step 1 Common multiple of 2 and 3: 6.

Step 2 Mix A $\times 3 = 6 : 15$. Mix B $\times 2 = 6 : 14$.

Step 3 Compare water: $14 < 15$.

Answer Mix B is stronger (less water for 6 soap)

WORDS common multiple (múltiplo común) • least common multiple (mínimo común múltiplo) • stronger / weaker (más fuerte / más débil)

PRACTICE

show your work • label units

1. Scale each ratio so the first term is 18.

a. $3 : 4 = 18 : \underline{\hspace{2cm}}$

b. $2 : 7 = 18 : \underline{\hspace{2cm}}$

c. $6 : 5 = 18 : \underline{\hspace{2cm}}$

d. $9 : 10 = 18 : \underline{\hspace{2cm}}$

2. **Purple Paint.** Mix A uses 4 cups red to 5 cups blue. Mix B uses 3 cups red to 4 cups blue. Complete the table. Which mix is bluer?

Hint Make the red 12 cups in both mixes.

	Red (cups)	Blue (cups)
Mix A $\times 3$	12	
Mix B $\times 4$	12	

Bluer mix _____

3. **Orange Juice.** Pitcher 1 has 4 cans of juice to 7 cans of water. Pitcher 2 has 3 cans of juice to 5 cans of water.

Hint Scale both pitchers to 12 cans of juice.

a. Scale both to 12 cans: _____

b. Which is more orange? _____

Show your work.

4. **Plant Food.** Bottle M mixes 3 drops of plant food with 8 ounces of water. Bottle N mixes 5 drops with 13 ounces.

a. Scale both to 15 drops: _____

b. Which is stronger? _____

Show your work.

- 5. Hot Cocoa.** Cup A has 3 scoops of cocoa in 10 oz of milk. Cup B has 2 scoops in 7 oz. Scale both to 6 scoops. Which cup is more chocolatey?

Hint $A \times 2$ and $B \times 3$.

Show your work.

Cup _____

- 6. Pool Chlorine.** Pool 1 uses 7 tablets for 10,000 gallons. Pool 2 uses 3 tablets for 5,000 gallons. Which pool has more chlorine per gallon?

Hint Scale Pool 2 up to 10,000 gallons.

Show your work.

Pool _____

- 7. Bubble Solution.** Jar 1: 2 spoons of soap to 5 cups of water. Jar 2: 3 spoons to 8 cups. Scale both to 6 spoons of soap. Which is stronger?

Show your work.

Jar _____

- 8. Team Wins.** Team 1 wins 3 of every 8 games. Team 2 wins 4 of every 10 games.

a. Scale both to 12 wins: _____

b. Which team wins a larger share of its games? Why? _____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

- 9. Lucas's Work.** Mix A: 3 cups red to 8 cups blue. Mix B: 4 cups red to 10 cups blue. Lucas says B is bluer because $10 > 8$. Explain his mistake. Which mix is bluer?

TEST PRACTICE

choose the best answer

- 10.** Scaled so the first term is 12, the ratio 3 : 8 becomes –

A 12 : 17

B 12 : 32

C 12 : 24

D 36 : 8

- 11.** Two cocoa mixes have the same amount of cocoa. The stronger mix has –

A more milk

B less milk

C more cups

D the same milk

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

12 : 7 Z	63 E	12 : 32 S	23 J	15 Q	12 : 21 U	6 : 16 L	15 : 40 A	15 : 20 X
#1b		#1c	#3a	#4a	#7	#8a		

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #____